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PERSONAL SKILLS

Profile

Applied artificial intelligence and knowledge management specialist for higher education institutions and industry, with over 10 years of professional experience. His experience includes participation in the SIRO project, six years of technical leadership of the CREAMINKA ecosystem, AI-based knowledge management for the research ecosystem at Pontificia Universidad Católica del Ecuador, and the development of intelligent modules for technology scouting in the Spanish industry within the framework of the University of Vigo. Throughout this professional path, he has built and maintained production systems based on ontologies, language models, embeddings, and knowledge modeling. He has also published more than 20 scientific articles in high-impact journals and international conferences. Academically, he is pursuing a PhD in Information and Communication Technologies at the University of Vigo. He holds a Master's Degree in Visual Analytics and Big Data from the International University of La Rioja, and a Systems Engineering degree from Universidad Politécnica Salesiana. His most recent lines of work, focused on governance-aware entity deduplication, semantic recommendation, and topic analysis with LLMs, reflect his interest in developing AI solutions that are not only accurate, but also explainable, governable, and useful for decision-making.

Organizational skills

Social responsibility, adaptability, change leadership, crisis management and problem solving, and social awareness leveraged through technology.

Technical skills

Advanced knowledge of ontologies, large language models (LLMs), embeddings, prompt engineering, text classification algorithms, clustering, and Social Network Analysis (SNA). Advanced knowledge of statistical languages (R, Python), business intelligence software (Tableau, Power BI, Data Studio), programming languages (Java, Python, JavaScript, R), databases (Oracle, PostgreSQL, MongoDB, GraphDB, Firestore, SQLite), and office software (Excel, Word, PowerPoint, Visio).

Languages

Spanish (Native language)
English B1 level (Cambridge PET)

PROFESSIONAL EXPERIENCE

February 2024 – December 2025

Vice-Rectorate for Research, Outreach and Innovation Research Manager

Pontificia Universidad Católica del Ecuador

Knowledge management and development of intelligent tools for Research Ecosystems.

January 2024 – December 2024	Technology portfolio management project with scouting tools University of Vigo – Regional Companies Development of intelligent modules applied to technology scouting for the Spanish industry using bibliometrics and language models.
October 2018 – December 2023	Vice-Rectorate for Research Research Assistant Universidad Politécnica Salesiana Knowledge management and development of intelligent tools for Research, Innovation and Entrepreneurship Ecosystems.
May 2017 – December 2023	CREAMINKA Project: An intelligent ecosystem for knowledge management, information discovery, and decision support in research Universidad Politécnica Salesiana CREAMINKA is a project of the UPS university community designed as an integrated research information management system. Its main objective is to collect and process research-related data, including actors, research groups, research lines, and dissemination channels, using advanced data mining and ontology techniques.
October 2016 – September 2018	Research Group in Artificial Intelligence and Assistive Technologies – Universidad Politécnica Salesiana Research Assistant Universidad Politécnica Salesiana Research on the scientific production of Universidad Politécnica Salesiana and knowledge modeling of the university research environment supported by ontologies.
October 2015 – June 2017	SIRO Project developed by the Research Group in Artificial Intelligence and Assistive Technologies and Fundación CIMA Developer Universidad Politécnica Salesiana Development of a web platform for managing therapists at Fundación CIMA in Cuenca, collecting data from tools used in therapies, and managing patient therapy processes using ontologies.

EDUCATION AND TRAINING

October 2023 – Present	PhD in Information and Communication Technologies University of Vigo
October 2018 – November 2019	Master's Degree in Visual Analytics and Big Data International University of La Rioja

April 2022 – September 2022	Online Diploma in BPM for Digital Process Transformation Pontificia Universidad Javeriana
September 2011 – February 2017	Systems Engineer with a specialization in Management Information Systems Universidad Politécnica Salesiana Top graduate of the 2011-2012 cohort of the Systems Engineering program. Vice President of the Student Federation of Universidad Politécnica Salesiana – Cuenca Campus, 2015-2017. President of the Systems Engineering program, 2015-2017. First place in the pure programming contest, master category, 2016. First place in the project contest, advanced category, 2014. First place in the programming contest at Universidad Politécnica Salesiana – Cuenca Campus, 2014. First place in the project contest, basic category, 2013. First place in the Phoenix National Programming Contest, Universidad de Cuenca, 2013. First place in the programming contest at Universidad Politécnica Salesiana – Cuenca Campus, 2013.

COURSES, SEMINARS, AND CONFERENCES

International presentations in Applied Artificial Intelligence (LLMs, Ontologies, and Knowledge Modeling)

May 2025 Lima, Peru	Speaker for “From manual processes to integrated digital ecosystems in research” at the Experiencias CRIS 2025 Conference PURE – Elsevier
June 2023 Lima, Peru	Speaker for “Discovering innovative collaborations through metadata: an embeddings- and LLM-based approach to leverage Pure and other bibliographic databases” at the Pure LATAM 2023 Conference PURE - Elsevier https://www.elsevier.com/es-mx/events/conferences/conferencia-pure-latam-2023
March 2025 Florianópolis, Brazil	Speaker for “Topic Analysis in News About AI: An Approach Based On LDA and Large Language Models” at the WorldCIST 2025 Conference
October 3-5, 2018 Bogotá, Colombia	Speaker for the paper “An intelligent system based on genetic algorithms to generate study groups using personality traits and academic profiles in higher education” and the paper “Integration of data based on ontologies for the generation of a multimedia repository of language and speech” at the 4th International Congress on Innovation and Trends in Engineering, CONIITI. Universidad Católica de Colombia
September 20-22, 2017 Cali, Colombia	Speaker for the paper “CREAMINKA: An intelligent ecosystem based on ontologies and artificial intelligence to manage research processes, knowledge generation and scientific production in higher education” at the Colombian Conference on Computing Sociedad Colombiana de Computación
June 15-17, 2016 Buenos Aires, Argentina	Speaker for the paper “An intelligent system based on ontologies and ICT tools to support the diagnosis and intervention of children with autism” at IEEE ARGENCON 2016 IEEE Argentina Section

Specialized training in AI, Machine Learning, and Cloud

April 2026 Online	Certificate in A Second Brain with Obsidian Coursera
June 2026 Online	Introduction to agent skills Anthropic Academy
March 2023 Cuenca, Ecuador	Course certificate: "Strategies for the Conservation and Promotion of Historical and Ethnographic Cultural Heritage. Perspectives from Technology, Citizenship, and Artificial Intelligence" CEDIA – Universidad Politécnica Salesiana – Universidad Católica de Cuenca
May – August 2018 Online	Deep Learning Nanodegree certificate Udacity, in collaboration with Amazon Web Services and Facebook Artificial Intelligence.
November 2020 Online	Course certificate: "Google Cloud Big Data and Machine Learning Fundamentals in Spanish" Coursera – Google Cloud
March 2020 Online	Course certificate: "Firebase for Web" Platzi
March 2020 Online	Course certificate: "Firebase Cloud Functions" Platzi
April 2026 Online	Certificate in Vibe Coding with GitHub Copilot Edureka
April 2026 Online	Google AI Fundamentals Google
April 2021 Online	Course certificate: "Natural Language Processing" Kaggle
April 2021 Online	Course certificate: "Intro to Machine Learning" Kaggle

Additional training in development and data analysis

May 2022 Online	ADONIS NP User Training BOC Group
July - August 2021 Online	Course certificate: "Data Science and Bibliometrics" Universidad Politécnica Salesiana
August 2020 Online	Course certificate: "R Fundamentals" Platzi
February 3, 4, 10, and 11, 2017 Cuenca, Ecuador	Participant in the course "Quantitative Research Methodology using R Software" Universidad Politécnica Salesiana
March 5-19, 2014 Cuenca, Ecuador	Seminar certificate: "Web Programming" Universidad Politécnica Salesiana
September 2013 – January 2014 Cuenca, Ecuador	Seminar certificate: "Mobile Applications" Universidad Politécnica Salesiana
March 2016 Cuenca, Ecuador	Seminar certificate: "CCNA Switching & Routing: Introduction to Networks" Universidad Politécnica Salesiana, Cisco Networking Academy

Training in management, innovation, and technology transfer

March 2021 Online	Course certificate: "Technological Innovation: From Strategy to Application" MIT xPRO
October 20-23, 2016, and April 24-25, 2017 Cuenca, Ecuador	Certificate of participation in the course "Knowledge and Technological Transfer" Universidad Politécnica Salesiana, NETVAL (Italian Network of Technology Transfer Offices of Universities and Public Research Organizations)
September 2021 Online	Course certificate: "Academic Social Networks" Universidad Politécnica Salesiana
December 2015 Cuenca, Ecuador	Certificate of participation in "Empower Project" Forum Ecuador
October 2014 Cuenca, Ecuador	Course certificate: "Google Apps Session 2014" Training Peru

SCIENTIFIC PUBLICATIONS

Galán-Mena, J., López-Nores, M., Pulla-Sánchez, D., Guerrero-Vásquez, L. F., & Salgado-Guerrero, J. P. (2026). From Research Spaces to Strategic Portfolio Design: Forecasting Country-Level Scholarly Diversification. *Mathematics*, 14(11), 1953. <https://doi.org/10.3390/math14111953>

Galán-Mena, J., López-Nores, M., Pulla-Sánchez, D., Guerrero-Vásquez, L. F., & Salgado-Guerrero, J. P. (2026). OntoDup: Governance-Aware Entity Matching for Scholarly Knowledge Graph Deduplication. *Information*, 17(4), 325. <https://doi.org/10.3390/info17040325>

Galán-Mena, J., López-Nores, M., Pulla-Sánchez, D., Guerrero-Vásquez, L. F., & Salgado-Guerrero, J. P. (2026). Unlikely Pairs: A Decision-Support Recommendation Pipeline for Discovering Semantically Plausible Research Collaborations. *Information*, 17(3), 254. <https://doi.org/10.3390/info17030254>

A. M. Pazmiño-Villamarín, M. López-Paredes, J. Cruz-Silva, J. Galán-Mena, D. Pulla-Sánchez and M. López-Nores, "Exploring Political Discourse in Ecuador's 2025 Election: Social Media Topic Modeling with BERTopic," 2025 IEEE Ninth Ecuador Technical Chapters Meeting (ETCM), Quito, Ecuador, 2025, pp. 1-6, doi: 10.1109/ETCM67548.2025.11304475.

J. Salazar-Pantusin, J. Galán-Mena, D. Pulla-Sánchez, M. López-Nores and J. P. Salgado-Guerrero, "Mentalization-Based Therapy Worldwide: A Bibliometric Mapping of Growth, Collaboration and Expansion Across Diagnoses," 2025 IEEE Ninth Ecuador Technical Chapters Meeting (ETCM), Quito, Ecuador, 2025, pp. 1-6, doi: 10.1109/ETCM67548.2025.11304388.

Galán-Mena, J., López-Nores, M., Galán, J., Pulla-Sánchez, D., Guerrero-Vásquez, L. F., & Salgado-Guerrero, J. P. (2025). Topic Analysis in News About AI: An Approach Based on LDA and Large Language Models. *Emerging Trends in Information Systems and Technologies: WorldCIST 2025 Volume 2*, 2, 355.

Neira-Maldonado, P., Quisi-Peralta, D., Salgado-Guerrero, J., Murillo-Valarezo, J., Cárdenas-Arichábal, T., Galan-Mena, J., & Pulla-Sanchez, D. (2024, January). Intelligent educational agent for education support using large language models through LangChain. In *International Conference on Information Technology & Systems* (pp. 258-268). Cham: Springer Nature Switzerland.

Quisi-Peralta, D. F., Lopez-Nores, M., Salgado, J. P., Murillo-Valarazo, J., Garcia-Velez, R., & Galan-Mena, J. (2024). Chatbot conversacional inteligente basado en la Comprensión del lenguaje natural y modelo de aprendizaje auto-regresivo GPT-3 para admisiones de la Universidad. *Revista Ibérica de Sistemas e Tecnologías de Informação*, (E66), 1-12.

Torres-Toukoumidis, A., Calle-Pesántez, S. E., Castro, S., & Galán-Mena, J. (2023, October). The Role of Twitter in Media Coverage during Humanitarian Crises. *Data mining from International News Agencies*. In *Conference on Information and Communication Technologies of Ecuador* (pp. 83-94). Cham: Springer Nature Switzerland.

León, R. A. P., Maldonado, F. S. C., Palomeque, F. L. B., & Mena, J. G. (2020). Reconocimiento de tres patrones básicos de movimiento de la mano utilizando electromiografía de superficie y algoritmos inteligentes. *Revista Cubana de Investigaciones Biomédicas*, 39(2).

Quisi-Peralta, D., Robles-Bykbaev, V., Galan-Mena, J., & García-Vélez, R. (2020). Automatic Generation of Abstracts in Scientific Articles Based on Natural Language Processing for Early Education Professionals and Speech Therapists. *International Conference on Applied Human Factors and Ergonomics*, 258–263.

Galán-Mena, J. A. (2019). Comparativa de técnicas para el descubrimiento de conocimiento aplicadas al comportamiento de producción científica.

Robles-Bykbaev, V., Galán-Mena, J., López-Nores, M., Robles-Bykbaev, Y., Cornejo-Reyes, J., Quisi-Peralta, D., ... León-Wong, V. C. (2018). A hybrid system to support the psychological intervention for children with autism: A proposal based on knowledge modeling and intelligent technologies. *International Journal of E-Services and Mobile Applications (IJESMA)*, 10(4), 74–100.

García-Vélez, R., Galán-Mena, J., López-Nores, M., & Robles-Bykbaev, V. (2018). Creating an Ontological Networks to Support the Inference of Personality Traits and Facets. 2018 IEEE XXV International Conference on Electronics, Electrical Engineering and Computing (INTERCON), 1–4.

García-Vélez, R., Galan-Mena, J., Robles-Bykbaev, V., López-Nores, M., Pesántes-Avilés, F., & Calle-López, D. (2018). An academic system based on ontological networks to support the inference of new knowledge in micro and macro curriculum in higher education. *International Conference on Applied Human Factors and Ergonomics*, 316–328.

Ingavélez-Guerra, P., Ulloa-Amaya, M., Vera-Rea, A., Galán-Mena, J., Robles-Bykbaev, V., Hilera, J. R., & Otón, S. (2018). An ontological network to identify accessibility metadata in learning objects: an approach based on Web Content Accessibility Guidelines, schemas, and disabilities analysis. 2018 IEEE International Autumn Meeting on Power, Electronics and Computing (ROPEC), 1–6.

García-Vélez, R., Robles-Bykbaev, V., López-Nores, M., Calle-López, D., Barros-Pontón, M., & Galán-Mena, J. (2018). An intelligent system based on genetic algorithms to generate study groups using personality traits and academic profiles in higher education. 2018 Congreso Internacional de Innovación y Tendencias En Ingeniería (CONIITI), 1–6.

Quisi-Peralta, D., Galan-Mena, J., López-Nores, M., Chaglla-Rodriguez, L., & Robles-Bykbaev, V. (2018). Integration of Data Based on Ontologies for the Generation of a Multimedia Repository of Language and Speech. 2018 Congreso Internacional de Innovación y Tendencias En Ingeniería (CONIITI), 1–6.

Guerrero, J. P. S., Sanchez, D. J. P., Mena, J. A. G., Bykbaev, V. E. R., Wong, P. V. C. L., & Narváez-Pacheco, A. (2018). CREAMINKA: An Intelligent Ecosystem for Supporting Management and Information Discovery in Research and Innovation Fields in Universities. In *Management of Information Systems* (p. 223). BoD--Books on Demand.

Olaleye, S. A., Salo, J., Sanusi, I. T., Hajiheydari, N., Hernandez-Carrion, J. R., Ashkani, M., ... others. (2018). *International Journal of E-Services and Mobile Applications*. *International Journal*, 10(4).

Robles-Bykbaev, V., Galán-Mena, J., López-Nores, M., Robles-Bykbaev, Y., Cornejo-Reyes, J., Quisi-Peralta, D., ... León-Wong, V. C. (2018). A Hybrid System to Support the Psychological Intervention for Children With Autism: A Proposal Based on Knowledge Modeling.

García-Vélez, R. A., Galán-Mena, J. A., Dahroug, A., Robles-Bykbaev, V. E., & López-Nores, M. (2018). Semantic representation of information by ontological networks to improve knowledge management in higher education.

Salgado-Guerrero, J. P., Galán-Mena, J., Pulla-Sánchez, D., Robles-Bykbaev, V., & Narváez-Pacheco, A. (2017). CREAMINKA: An intelligent ecosystem based on ontologies and artificial intelligence to manage research processes, knowledge generation and scientific production in higher education. *Colombian Conference on Computing*, 115–129.

Robles-Bykbaev, V., López-Nores, M., Galán-Mena, J. A., Wong, V. C. L., Quisi-Peralta, D., Lima-Juma, D., ... Pazos-Arias, J. J. (2017). An intelligent ecosystem to support the development of communication skills in children with autism: An experience based on ontologies, multi-sensory stimulation rooms, and robotic assistants. In *Smart Technology Applications in Business Environments* (pp. 109–133). IGI Global.

Galán-Mena, J., Ávila, G., Pauta-Pintado, J., Lima-Juma, D., Robles-Bykbaev, V., & Quisi-Peralta, D. (2016). An intelligent system based on ontologies and ICT tools to support the diagnosis and intervention of children with autism. 2016 IEEE Biennial Congress of Argentina (ARGENCON), 1–5.

ENTREPRENEURIAL EXPERIENCE

June 2020	Technical team member for the reCREATE 2020 event Universidad Politécnica Salesiana
	Web platform developer for the Entrepreneurship Ecosystem https://coworking.ups.edu.ec Universidad Politécnica Salesiana
May 2019	Technical team member for the rETHOS 2019 event Universidad Politécnica Salesiana
April 2018	Technical team member for the rETHOS 2018 event Universidad Politécnica Salesiana